Addition and subtraction of radicals

1 Simplify the following:

a $8\sqrt{3}+18\sqrt{3}$

b $-12\sqrt{5}+6\sqrt{5}-12\sqrt{5}$

c $8\sqrt{6}+17\sqrt{13}+19\sqrt{6}-6\sqrt{13}$

d $17\sqrt{6}-8\sqrt{11}+3\sqrt{6}+13\sqrt{11}$

e $\sqrt{6}+\sqrt{54}$

f $\sqrt{180}+\sqrt{500}$

g $\sqrt{32}-\sqrt{98}$

h $4\sqrt{45}+5\sqrt{180}$

i $\sqrt{10}+10\sqrt{10}$

j $-19\sqrt{5}+4\sqrt{5}$

k $18\sqrt{2}-10\sqrt{2}+20\sqrt{2}$

l $19\sqrt{3}+19\sqrt{7}-7\sqrt{7}+22\sqrt{3}$

m $\sqrt{243}+\sqrt{3}$

n $2\sqrt{75}-3\sqrt{108}$

o $\dfrac{12+9\sqrt{5}}{3}$

2 Use the formula $\sqrt{a}+\sqrt{b}=\sqrt{a+b+2\sqrt{ab}}$ to simplify the following:.

a $\sqrt{3}+\sqrt{12}$

b $\sqrt{8}+\sqrt{2}$

c $\sqrt{18}+\sqrt{8}$

d $2\sqrt{20}+2\sqrt{5}$

3 Simplify the following:

a $\sqrt[3]{512v}-5\sqrt[3]{v}$

b $2\sqrt[3]{u}+\sqrt[3]{125u}$

c $\sqrt{ax^5}+x^2\sqrt{ax}$

Multiplication and division of radicals

4 Simplify the following:

a $\sqrt{2}\times\sqrt{11}$

b $2\sqrt{2}\times 2$

c $4\sqrt{35}\times\sqrt{5}$

d $5\sqrt{7}\times 3\sqrt{5}$

e $\sqrt{5}\times\sqrt{2}\times\sqrt{7}$

f $\sqrt{343}\times\sqrt{75}$

g $\sqrt{5}\times\sqrt{7}$

h $8\times 10\sqrt{5}$

i $\sqrt{7}\times\sqrt{3}\times\sqrt{11}$

j $\sqrt{55}\times\sqrt{11}$

k $4\sqrt{11}\times 5$

l $2\sqrt{5}\times 15\sqrt{11}$

m $7\sqrt{22}\times\sqrt{2}$

n $\sqrt{180}\times\sqrt{48}$

o $8\sqrt{15}\times 8\sqrt{5}$

p $5\sqrt{17}\times 8\sqrt{3}$

q $17\sqrt{35}\times 4\sqrt{5}$

r $8\sqrt{51}\times 9\sqrt{3}$

5 Simplify the following:

a $\sqrt{85}\div\sqrt{17}$

b $18\sqrt{7}\div 9$

c $24\sqrt{112}\div 3\sqrt{7}$

d $8\sqrt{77}\div\sqrt{7}$

e $20\sqrt{14}\div 5\sqrt{2}$

f $\sqrt{15}\div\sqrt{5}$

g $\sqrt{55}\div\sqrt{5}$

h $\sqrt{51}\div\sqrt{17}$

i $\sqrt{21}\div\sqrt{3}$

j $\sqrt{91}\div\sqrt{7}$

k $40\sqrt{7}\div 8$

l $10\sqrt{55}\div\sqrt{11}$

m $15\sqrt{22}\div\sqrt{11}$

n $4\sqrt{35}\div 2\sqrt{5}$

o $\sqrt{27}\div\sqrt{3}$

p $3\sqrt{20}\div\sqrt{5}$

q $5\sqrt{8}\div\sqrt{2}$

r $40\sqrt{96}\div 10\sqrt{6}$

s $50\sqrt{24}\div 10\sqrt{6}$

t $\sqrt{25}\div\sqrt{81}$

u $\sqrt{162}\div\sqrt{8}$

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6 Simplify the following:

a $\sqrt{\frac{28}{7}}$ b $\sqrt{\frac{9}{45}}$ c $\sqrt{\frac{64}{4}}$ d $\sqrt{\frac{48}{144}}$

e $\frac{\sqrt{12}}{\sqrt{36}}$ f $\frac{\sqrt{56}}{\sqrt{14}}$ g $\frac{\sqrt{36}}{\sqrt{81}}$ h $\frac{\sqrt{72}}{\sqrt{32}}$

i $\frac{17\sqrt{21}}{\sqrt{3}}$

7 Assuming that all variables are non-negative, simplify the following:

a $\sqrt{4j}\sqrt{j}\sqrt{4k}$ b $y\sqrt{y}y\sqrt{y}$

8 Simplify the following:

a $\frac{\sqrt[3]{6}}{\sqrt[3]{6 \times 7^3}}$ b $\frac{\sqrt{2p}}{\sqrt{2q}}$ c $6\sqrt{7m} \div \sqrt{4m}$

Mixed operations with radicals

9 Distribute and simplify the following:

a $\sqrt{7}\left(-5+\sqrt{5}\right)$ b $7\sqrt{2}\left(\sqrt{7}+6\right)$

c $7\sqrt{11}\left(-6\sqrt{2}+\sqrt{7}\right)$ d $\sqrt{11}\left(\sqrt{7}+4\right)$

e $\sqrt{7}\left(3+\sqrt{3}\right)$ f $\sqrt{2}\left(\sqrt{11}-6\right)$

g $3\sqrt{3}\left(\sqrt{13}-5\right)$ h $\sqrt{3}\left(\sqrt{11}+\sqrt{13}\right)$

i $4\sqrt{7}\left(\sqrt{2}-\sqrt{11}\right)$ j $3\sqrt{5}\left(\sqrt{55}+\sqrt{11}\right)$

k $8\sqrt{2}\left(\sqrt{3}-3\sqrt{7}\right)$ l $5\sqrt{2}\left(3\sqrt{5}+4\sqrt{7}\right)$

m $7\sqrt{3}\left(\sqrt{15}+\sqrt{60}\right)$ n $11\sqrt{3}\left(3\sqrt{5}-\sqrt{20}\right)$

o $8\sqrt{11}\left(3\sqrt{7}-4\sqrt{5}\right)$ p $\sqrt{13}\left(\sqrt{11}+2\right)$

q $8\sqrt{11}\left(-3\sqrt{5}-7\sqrt{7}\right)$

10 Distribute and simplify the following:

a $\left(\sqrt{13}+5\right)\left(\sqrt{2}-3\right)$ b $\left(\sqrt{7}-\sqrt{13}\right)\left(\sqrt{2}-\sqrt{11}\right)$

c $\left(\sqrt{11}+1\right)^2$ d $\left(2\sqrt{5}+3\right)^2$

e $\left(5-\sqrt{7}\right)\left(8-\sqrt{3}\right)$ f $\left(3\sqrt{11}-\sqrt{5}\right)^2$

g $\left(\sqrt{11}-11\right)\left(\sqrt{11}+11\right)$ h $\left(\sqrt{10}+\sqrt{7}\right)\left(\sqrt{10}-\sqrt{7}\right)$

i $\left(4\sqrt{2}-\sqrt{7}\right)\left(4\sqrt{2}+\sqrt{7}\right)$

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11 Simplify the following:

a $3\sqrt{5} \times 8\sqrt{10} \div 2\sqrt{2}$

b $\dfrac{8\sqrt{12} \times 3\sqrt{2}}{4\sqrt{10}}$

c $\dfrac{\sqrt{15} + \sqrt{21}}{\sqrt{3}}$

d $\dfrac{3\sqrt{2} \times 2\sqrt{6}}{\sqrt{8} \times \sqrt{20}}$

12 Find the values of x and y in the following equations:

a $\left(8\sqrt{91} - 5\sqrt{7}\right)^2 = x + y\sqrt{13}$

b $\left(\sqrt{x} + y\right)^2 = 14 - 6\sqrt{5}$

13 Consider the expression $\left(\sqrt{13} + 4\right)^2 + \left(\sqrt{13} + m\right)^2$.

a Distribute and simplify the expression.

b What value of m can be substituted into $\left(\sqrt{13} + 4\right)^2 + \left(\sqrt{13} + m\right)^2$ so that it has a rational value?

14 Simplify the following:

a $\sqrt[3]{u^2}\sqrt[3]{u}$

b $\sqrt[3]{3u^2}\sqrt[3]{9u}$

c $7\sqrt{u} + 3\sqrt{u}$

d $10\sqrt{u} - 8\sqrt{v} - 4\sqrt{u} + 9\sqrt{v}$

e $6\sqrt{a} + \sqrt{81a}$

f $\sqrt{125a} - \sqrt{20a}$

Rationalize the denominator

15 State whether the following results in a rational number:

a $\left(\sqrt{8} + \sqrt{3}\right) \times \left(\sqrt{8} + \sqrt{3}\right)$

b $\left(\sqrt{8} + \sqrt{3}\right) \times \left(\sqrt{8} - \sqrt{3}\right)$

c $\left(-\left(\sqrt{8} + 2\right)\right) \times \sqrt{8}$

16 Rationalize the denominator of the following:

a $\dfrac{1}{\sqrt{11}}$

b $\dfrac{\sqrt{2} - 2}{\sqrt{11}}$

c $\dfrac{7}{\sqrt{12}}$

d $\dfrac{14\sqrt{11}}{9\sqrt{13}}$

e $-\dfrac{3}{5\sqrt{13}}$

f $\dfrac{7\sqrt{30}}{\sqrt{10}}$

g $\dfrac{10\sqrt{7} - 1}{\sqrt{5}}$

h $\dfrac{\sqrt{5}}{\sqrt{15}}$

17 Find a conjugate for the following expressions:

a $2 + \sqrt{7}$

b $8 - \sqrt{u}$

c $4 - 3\sqrt{10}$

d $-5\sqrt{3} + 4\sqrt{v}$

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18 Rationalize the denominator for the following expressions:

a $\dfrac{9}{\sqrt{5}-7}$ **b** $\dfrac{28}{\sqrt{10}-\sqrt{3}}$ **c** $\dfrac{\sqrt{7}}{5\sqrt{6}+4\sqrt{2}}$ **d** $\dfrac{\sqrt{11}+\sqrt{2}}{\sqrt{11}-\sqrt{2}}$

e $\dfrac{9\sqrt{7}+8\sqrt{5}}{9\sqrt{7}-8\sqrt{5}}$ **f** $\dfrac{8}{9\sqrt{11}-5}$ **g** $\dfrac{\sqrt{7}+\sqrt{2}}{\sqrt{7}-\sqrt{2}}$ **h** $\left(\sqrt{10}+1\right)^{-1}$

19 Express the following in simplest form with rational denominator:

a $\dfrac{3}{\sqrt{10}}+\dfrac{1}{4}$ **b** $\dfrac{3}{4\sqrt{5}}-\dfrac{3}{2\sqrt{2}}$

c $\dfrac{5\sqrt{3}}{6\sqrt{5}}-\dfrac{7\sqrt{5}}{3\sqrt{3}}$ **d** $\dfrac{1}{\sqrt{2}+5}+\dfrac{1}{\sqrt{2}-5}$

20 Show that the following expression is a rational number:

$$\frac{3}{\sqrt{2}+3}+\frac{3}{\sqrt{2}-3}$$

21 If $x=2+\sqrt{3}$, simplify and rationalize the denominator of the following expressions:

a $\dfrac{x-\sqrt{3}}{x+\sqrt{3}}$ **b** $\left(x+\dfrac{1}{x}\right)^2$

22 If $x=4+\sqrt{3}$, simplify and rationalize the denominator of the following expressions:

a x^2 **b** $\dfrac{x+1}{x-1}$

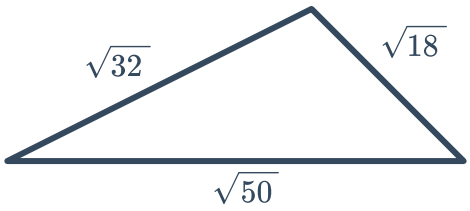
23 Express the following as a single radical:

$$42+\frac{6}{7+5\sqrt{2}}$$

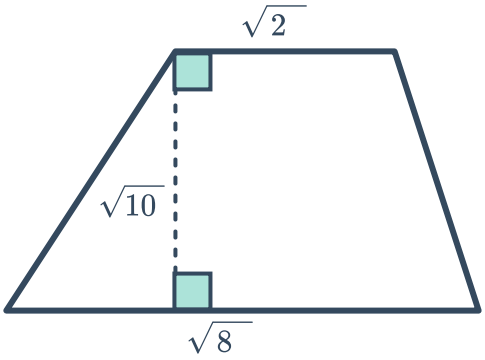
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Applications

24 Find the perimeter of the triangle in simplified radical form:



25 Find the area of the trapezoid in radical form:

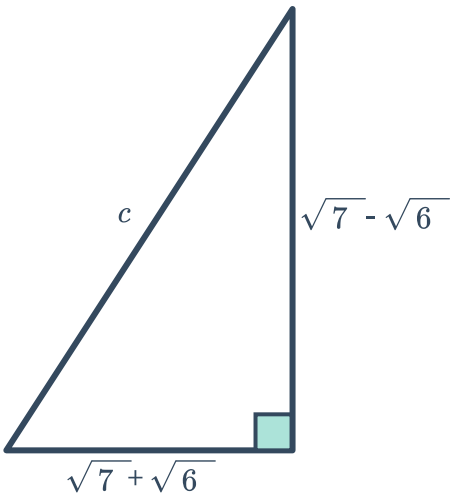


26 A rectangle has a width of $\left(\sqrt{7}-2\right)$ ft and a length of $\left(\sqrt{7}+2\right)$ ft.

a Find the exact perimeter of the rectangle.

b Find the exact area of the rectangle.

27 Find c, the length of the hypotenuse of the triangle below. Leave your answer in radical form.



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28 Find the exact perpendicular height of a triangle whose area is $36\sqrt{15}$ square centimeters and whose base measures $8\sqrt{5}$ centimeters.

29 A rectangle with height h and width w is believed to be more aesthetically pleasing if its dimensions are in the following ratio:

$$\frac{w}{h} = \frac{2}{\sqrt{5}-1}$$

- **a** Rationalize the denominator of the ratio.
- **b** Find a decimal approximation for this ratio correct to the nearest thousandth.

30 A rectangle has a length of $\left(\sqrt{10}+3\right)$ meters and an area of 4 square meters. What is the exact width of this rectangle? Give your answer with a rational denominator.

31 The formula $P = \dfrac{x\left(12+\sqrt{x}\right)}{5\sqrt{x}}$ models the percentage, P, of bees that do not survive relocation to a new hive in a colony that has x thousand bees.

- **a** In a hive with $25\,000$ bees, what percentage will not survive relocating to a new hive? Round your answer to the nearest integer.
- **b** Rewrite the formula for P by rationalizing the denominator.
- **c** Using the formula you found in part (b), find the percentage of bees from a hive with $25\,000$ bees that will not survive relocation. Round your answer to the nearest integer.

32 Consider a square pyramid with a height h and a volume V. The length of one side of its base is given by $\sqrt{\dfrac{3V}{h}}$.

- **a** Rewrite the expression for the length such that it has a rational denominator.
- **b** Find the side length of the base of a square pyramid that has a height of 3 inches and a volume of 16 cubic inches.

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